Nvidia Test   NITk _Surathkal ..Date : 18/09/06 (Visited on)   K PAVAN

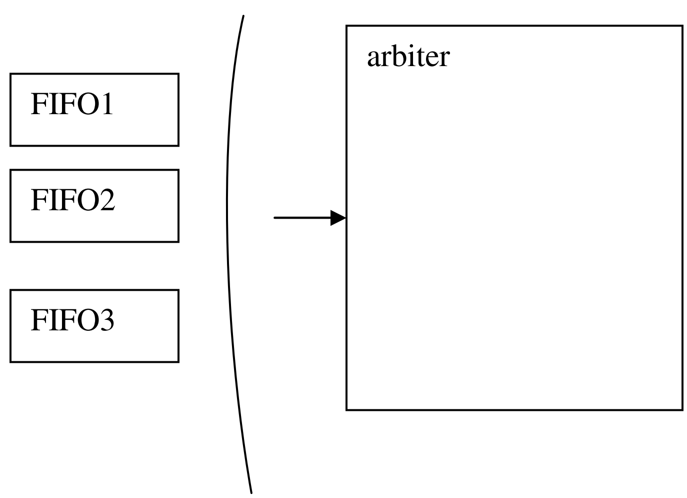
Descriptive, 1 hour , % questions.
Common paper for computers and E&C people.

Marks distribution ➔ 8,10,14,8,10.

Each question consists of at least 2 subsections.

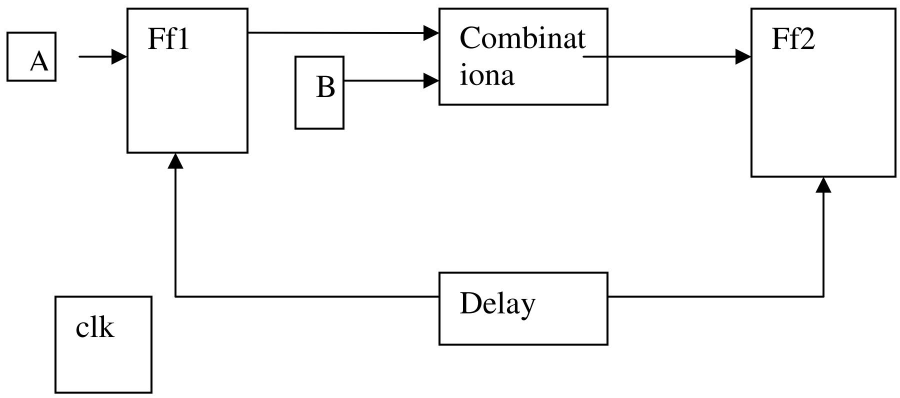
I did not read the entire question paper. But I found first 2 r 3 questions are too lengthy to read also. Thattoo they related to puzzles kindof stuff.
They told to attempt as many as you can and not to stuck at a single question.
You have to give analysis ,,how did u solve that problem along with any assumptions u made.
I started from 5th question.

5 a) problem based on FIFO.



Roughly it is shown in above figure.
Each FIFO has i.p data rates are given. Arbiter is just like a sampler which samples just 1 byte per one clock from each FIFO in Round robin method.
i.p data rates I am not sure. Like 6/30,14/30,8/20 like that.
Find depth of each FIFO.

5 b)
Question on finding max freq of operation .



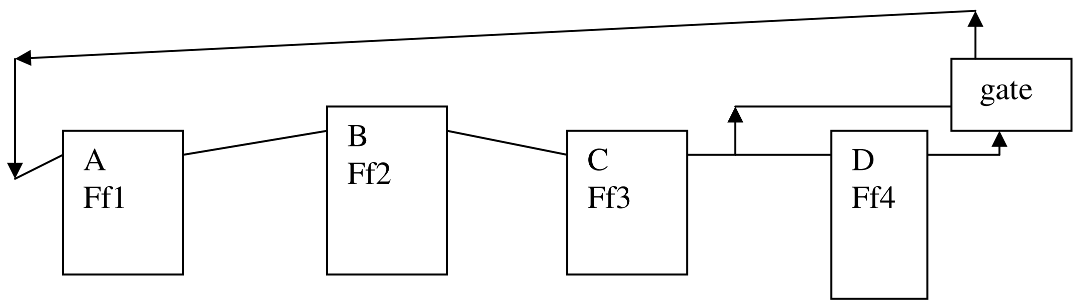
For FF1 and FF2 difference setup/hod time/ clk to Q delays are given.
I/p A comes from o/p of type 2 FF
I/p B comes through o/p of type 1 FF and through some 4.2 ns combinational delay.

This data is given.

I think the path from B->comb-> FF2 becomes critical path as B i/p comes from another FF1and some combinational delay. So with added skew on colk path calculate the max frequency.

Just check for any hold violation through shortest paths.

4 b) one sequential ckt given and asked to draw wave forms.
All ffs have same clear signal..it is a synchronous clear.
Clock is ssame form first 3 ffs. For 4 th ff the clock is the o/p of an xnor gate.
The both i/ps of x not gate are clock only. So clk for ff4 is one always.
All are +ve edge triggered flops. So data at ff4 is captured only at initiatl stages.



I don't remember the gate connected in feed back. Bit any way the o/p of ff4 is constant as its clock is set to one after some time. I thibnk it is xor gate. I found its value at o/p is c bar.

Finally it became.. a = cbar (without delay) ;b = delayed version of a , c = delayed version of b.

At one stage some synchronous set/reset is applied and asked to draw waveforms at A B C D ports..

4 a) Fibonaacci series .. 1 ,2 , 3 , 5 , 8, 13..

Represented by some 4 bit unsigned representation.

A,b,c,d are Boolean varialbles.. a is lsb. D is msb.

A function defined as F(a,b,c,d) = 1 if i/p is in Fibonacci series otherwise zero.

Find some bboolean expressions for a,b,c,d. (I remember upto this part only..i did not read this one properly..i think u r supposed to design some sequential circuit.)

3 b)

Draw the state machine….

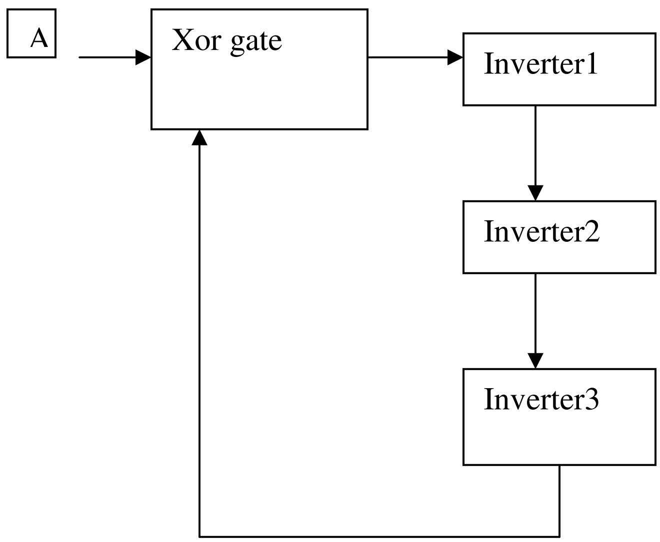
Which detects both 010 and 011 seqences. It should not take more than 5 states.

My frns tols they got it in 2 r 3 states..i didn't find time to do this. When I came upto this question only five minutes left.

One 14 marks question consist of.. 14 one mark questions..so I tried that one.

Some questions I remember are….

1) diff b/w trap and interrupt
2) multiprocessing is advantageous that threads?? Explain
3) howmany 4:1 mux required to build 64:1 mux
4) one question on number system…not sure..
    find a and b satisfying condition.. a>b && a-b < 0
    a,b are in 2's complement form ..32 bit numbers.
    My frns told answer as.. a = '1'all zeros and '1' all ones…
5)



Circuit is combinational r seqential….give reason

Depending on value of a some times it become ring oscillator some times it become buffer. I think it is seqential.

6) synchronous melay machine is glitch free at o/p r not? Give reason.
    I think it produces gliteches at o/p. if o/p is registered then it wil become glitch free. Generally we deal with synchronous state machines only..so particularly specifying synchgronous machine..means..already o/p is also synchronized r what.. I have this doubt…

This is what I remember..i ddinot see first half of the paper. And it is full of puzzles .some what related to finding order of an algorithm..kind of stuff. Means. Answers amy be in the form of logn base 2 .etc.

For some problems they asked to find out simple techniques..one guy told the solution is based on some of searching technique…
Some one told first 3 questions are interlinked..
Finally they shortlisted 2 mtech guys..and they did not take any one. They reduced the cutoff to 7.5 gpa and allowed persons who have a gpa of above 7 for test on request.they took 3 btech guys.

Thank u ..all the best.. I wil send some more details if I come to know them…..
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